

Junsu Kim

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Research Interests

Computer Architecture, Systems for ML & ML for Systems

Education

University of British Columbia, Vancouver, BC, Canada

Sep. 2025 - Current

Ph.D. in Electrical and Computer Engineering (Advisor: Prof. Prashant Nair) Cumulative GPA: 4.0/4.0

Korea University, Seoul, Korea

Sep. 2023 - Aug. 2025

M.S. in Electrical Engineering (Advisor: Prof. Yunho Oh)

Cumulative GPA: 4.0/4.0

Hanyang University, Seoul, Korea

Mar. 2014 - Feb. 2021

B.S. in Electronic Engineering (Advisor: Prof. Ki-Seok Chung)

Cumulative GPA: 3.81/4.0 (Graduating with Honors - Summa Cum Laude)

Publications

Conference [C] and Journal [J] Papers

[J2] **Junsu Kim**, Jaebeom Jeon, Jaeyong Park, and Sangun Choi, Minseong Gil, Seokin Hong, Gunjae Koo, Myung Kuk Yoon, and Yunho Oh “Memory Oversubscription-Aware Tensor Migration Scheduling for GPU Unified Storage Architecture” The IEEE Computer Architecture Letters (CAL), 2025.

[C5] **Junsu Kim**, and Suhyun Kim, “Salient Frequency-aware Exemplar Compression for Resource-constrained Online Continual Learning”, The 39th Annual AAAI Conference on Artificial Intelligence (AAAI), 2025.

[C4] Seondeok Kim*, Sangun Choi*, Jaebeom Jeon, **Junsu Kim**, Minseong Gil, Jaehyeok Ryu, and Yunho Oh, “Kubism: Disassembling and Reassembling K-Means Clustering for Mobile Heterogeneous Platforms”, The 26th ACM SIGPLAN/SIGBED International Conference on Languages, Compilers, and Tools for Embedded Systems (LCTES), 2025.

[C3] Dongwon Yang, Jaebeom Jeon, Minseong Gil, **Junsu Kim**, Seondeok Kim, Gunjae Koo, Myung Kuk Yoon, and Yunho Oh, “SSFFT: Energy-Efficient Selective Scaling for Fast Fourier Transform in Embedded GPUs”, The 26th ACM SIGPLAN/SIGBED International Conference on Languages, Compilers, and Tools for Embedded Systems (LCTES), 2025.

[J1] Minseong Gil, Jaebeom Jeon, **Junsu Kim**, Sangun Choi, Gunjae Koo, Myung Kuk Yoon, and Yunho Oh, “TLP Balancer: Predictive Thread Allocation for Multi-Tenant Inference in Embedded GPUs”, The IEEE Embedded Systems Letters (ESL), 2024.

[C2] Jaebeom Jeon, Minsung Gil, **Junsu Kim**, Jaeyoung Park, Gunjae Koo, Myung-Kuk Yoon, and Yunho Oh. “VitBit: Enhancing Embedded GPU Performance for AI Workloads through Register Operand Packing”. The 53rd International Conference on Parallel Processing (ICPP), 2024

[C1] Kwangrae Kim, Jeonghyun Woo, **Junsu Kim**, and Ki-Seok Chung. “HammerFilter: Robust Protection and Low Hardware Overhead Method for RowHammer”. The 39th IEEE International Conference on Computer Design (ICCD), 2021

Preprints (Project Names Only)

[P1] Co-author, “Hardware Supports for Enabling Arbitrary Numeric Format on GPUs”

Work Experience

University of British Columbia, Vancouver, BC, Canada

Sep. 2025 - Current

Research Assistant at Systems and Architectures (STAR) Lab

Advisor: Prof. Prashant Nair

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Korea University, Seoul, Korea

Sep. 2023 - Aug. 2025

Research Assistant at Computer Architecture and System Software Lab (ComSys)

Advisor: Prof. Yunho Oh

Korea Institute of Science and Technology, Seoul, Korea

May. 2022 - Aug. 2023

Research Assistant at Korea Data Science Team (KDST)

Supervisor: Dr. Suhyun Kim

Hanyang University, Seoul, Korea

Dec. 2019 - Mar. 2020, Aug. 2020 - Nov. 2020

Research Assistant at Embedded System on Chip Laboratory (ESOC Lab)

Advisor: Prof. Ki-Seok Chung

Research Assistant at Computer Architecture and System SW Lab (CASS Lab)

Advisor: Prof. Yongjun Park

Teaching Experience

University of British Columbia, Vancouver, BC, Canada

Fall 2025

Teaching Assistant for Computer Architecture (CPEN 411)

Korea University, Seoul, Korea

Spring 2024, Fall 2024

Teaching Assistant for Computer Architecture

School for the Blind, Chuncheon, Korea

Mar. 2017 - Feb. 2019

Assistant Teacher (Alternative Military Service)

Honors and Awards

Faculty of Applied Science Graduate Award, University of British Columbia (UBC) → \$7200 CAD

Nov. 2025

AAAI Student Travel Grant → ~ \$1600 USD

Mar. 2025

Korea University Graduate School Scholarship → Half Tuition (~ \$10000 CAD)

Spring 2024, Fall 2024

Korea University Graduate School Scholarship for Outstanding New Student → Half Tuition (~ \$5000 CAD)

Fall 2023

Hanyang University Scholarship → Full Tuition (~ \$20000 CAD)

2014 - 2019

Research Projects

Accuracy-preserving Tiered Key Value (KV) Cache Management for Large Language Model (LLM)

Advisor: Prof. Prashant Nair, University of British Columbia

Sep. 2025 - Current

- ◇ Proposed a tiered KV cache management framework that leverages a full KV cache stored in host DRAM and a compressed KV cache residing in GPU HBM
- ◇ Proposed reforwarding that rejects a new output token generated by the compressed KV cache and generates a new one with full KV cache depending on output token confidence
- ◇ Proposed KV cache recomputation that recomputes KV cache generated by the compressed KV cache during reforwarding
- ◇ Reduced the KV cache size by 60% while achieving an average speedup of 65.4% on LongWriter (Long-generation) and Longbench-v1 (Long-context), with less than 1% accuracy loss.
- ◇ Contributions: first-author, motivation study, idea, implementation

Memory Oversubscription-aware Tensor Migration Scheduling for GPU Unified Storage Architecture [CAL]

Advisor: Prof. Yunho Oh, Korea University

Feb. 2024 - Sep. 2024

- ◇ Analyzed page faults due to memory oversubscription stalled AI workloads when expanding GPU memory with SSD using UVM
- ◇ Proposed a tensor migration scheduling algorithm considering GPU memory oversubscription for GPU unified storage architecture
- ◇ Achieved the averaged speedup by 12.9% compared to G10, which was presented at MICRO 2023
- ◇ Contributions: 1st author, motivation study, idea, implementation, experiment, paper write-up

Hardware Supports for Enabling Arbitrary Numeric Format on GPUs

Advisor: Prof. Yunho Oh, Korea University

May. 2024 - Nov. 2024

- ◇ Observed GPU supports a limited set of numeric formats, wasting register files when processing arbitrary numeric formats
- ◇ Employed bitslice representation, which transposes the data elements, packing arbitrary numeric formats without register wastage.
- ◇ Proposed Bitslice Vector multiplier and adder, constructing a tree structure to replace a multiplication-adder tree in a Tensor core
- ◇ Contributions: co-author, motivation study, idea, implementation, paper write-up

A Behavioral Analysis of CXL Memory Systems

Advisor: Prof. Yunho Oh, Korea University

Sep. 2023 - Sep. 2024

Collaborator: SK hynix

- ◇ Observed the behavior of a real CXL-based system on datacenter and AI workloads in the CXL-based platform
- ◇ Analyzed how the different promotion and demotion methods for CXL devices affected the performance of the workloads
- ◇ Presented performance modeling for datacenter workloads using different system factors (e.g., memory bandwidth, memory latency)
- ◇ Contributions: co-author, experiment, analysis, paper write-up

Salient Frequency-aware Exemplar Compression for Online Continual Learning [AAAI'25]

Supervisor: Dr. Suhyun Kim, Korea Institute of Science and Technology

Jan. 2023 - Nov. 2023

- ◇ Observed exemplar compression methods occupied limited GPU resources during online continual learning
- ◇ Proposed a computationally efficient compression algorithm using salient frequency
- ◇ Proposed a buffer management scheme to alleviate harmful effects from the compression artifacts remaining in the buffer
- ◇ Contributions: 1st author, motivation study, idea, implementation, paper write-up

Skills

C/C++, Python, Tensorflow, Pytorch, Git, Verilog, Shell script